Tour de Maths 2018

Leg 7

Question Paper

5 Mark Questions

Question 1

Find the smallest natural number which, when divided by 2, 3, 4, 5 and 6 respectively, leaves a remainder of 1 every time.

Question 2

The sum of two positive integers is 104. What is the difference between the greatest and least possible products of the two numbers?

Question 3

What is the first number in this pattern?

?	21420	3060	612	204

Question 4

In the magic square alongside, the sum of the three numbers in each row, column and diagonal adds up to 30.

What number comes in block A?

	A	$11\frac{2}{3}$
$18\frac{1}{3}$		
		$16\frac{2}{3}$

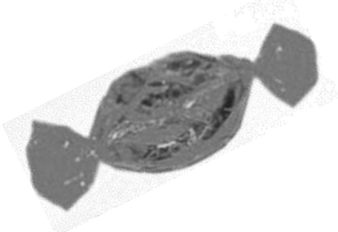
Question 5

If A, B and C are all positive integers, calculate the value of C if :

$A \times B = 48$, $A + B = 19$ and $A + C = 14$

Question 6

One toffee and one chocolate bar cost R9,00. Ten toffees and five chocolate bars cost R47,00.
How much does one chocolate bar cost?



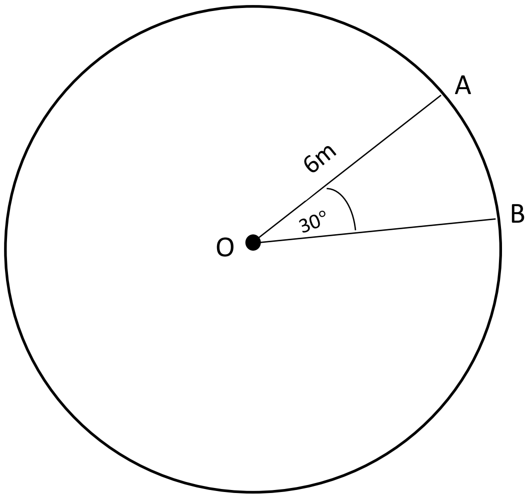
Question 7

A rectangular swimming pool is 3 times as long as it is wide. If it were 5 metres shorter and 3 metres wider, it would be square. What is the perimeter of the swimming pool?

Question 8

If $x + y + z = 28$, $x - y - z = 10$ and $z = 2$, then find the value of $2xyz$.

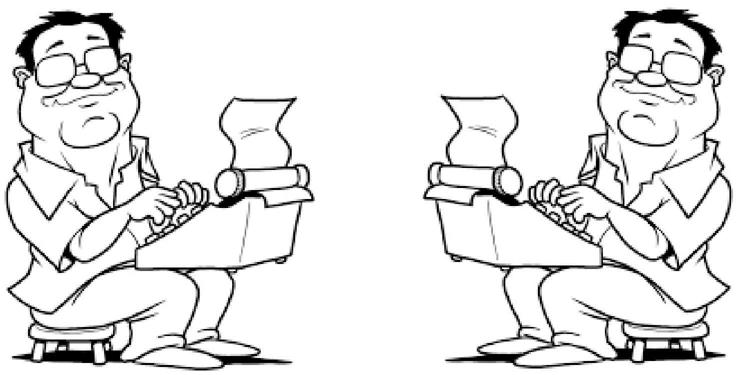
Question 9



In the figure, A and B lie on the circumference of the circle centred at O. OA is $6m$ long, and the measurement of $\angle AOB$ is $30°$. How long is the minor arc AB in terms of π?

Question 10

If two typists can type two pages in two minutes, how many typists will it take to type 18 pages in six minutes?



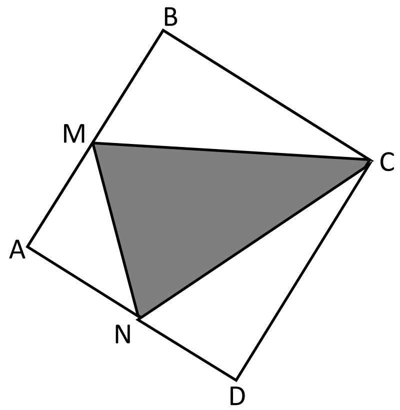
10 Mark Questions

Question 11

294 digits are used to number the pages of a book. How many pages does the book have?

Question 12

ABCD is a square.
M is halfway between A and B.
N is halfway between A and D.
What fraction is the shaded part of the square?



Question 13

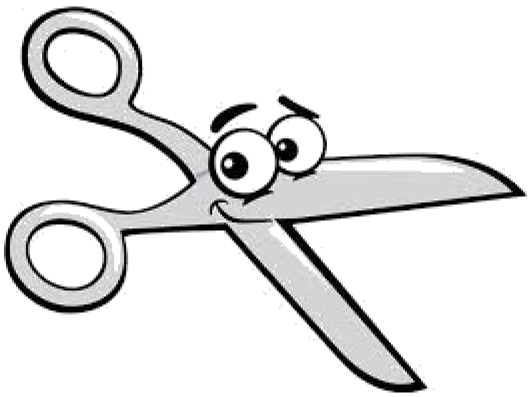
The price of a new pair of jeans is decreased by 20%. This new price is then increased by 30%. What percentage is the final price more than the original price?

Question 14

Today is Thursday, 25 October, 2018. In which year will the next October 25th fall on a Thursday?

Question 15

Bob cut a sheet of paper into 8 pieces. Then he took one piece and cut it again into 8 pieces. He then repeated this four more times. How many pieces of paper did he have after the last cutting?



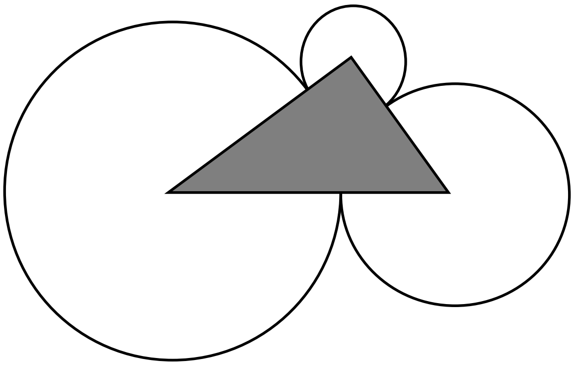
20 Mark Questions

Question 16

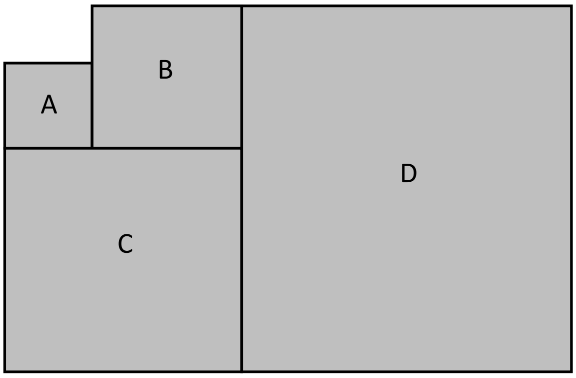
This morning at 6:00, my watch was 3 minutes slow, but at 13:00 it was 7 minutes fast!
At what time was it exactly correct?

Question 17

The vertices of the shaded triangle are at the midpoints of each circle. The dimensions of the sides of the triangle are 58, 44 and 36 units. Find the radius of the largest circle.



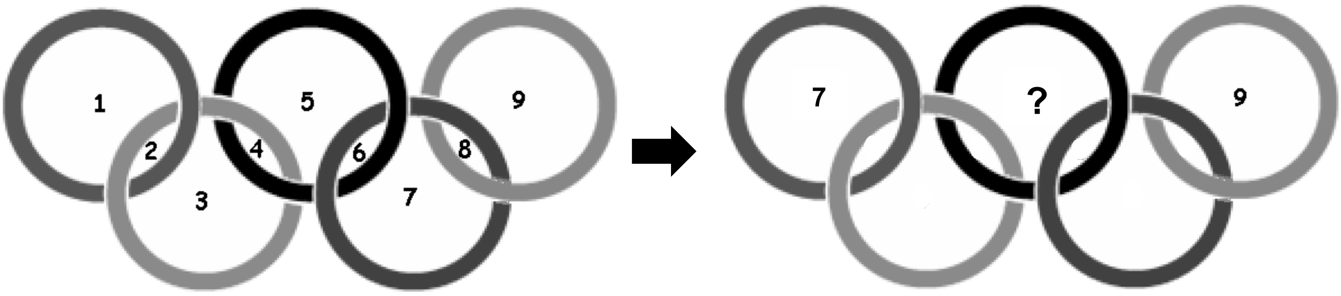
Question 18



The perimeter of Square A is $120cm$ and the area of Square B is $0{,}64m^2$. Figures C and D are also squares. What is the area of the entire shaded region in m^2?

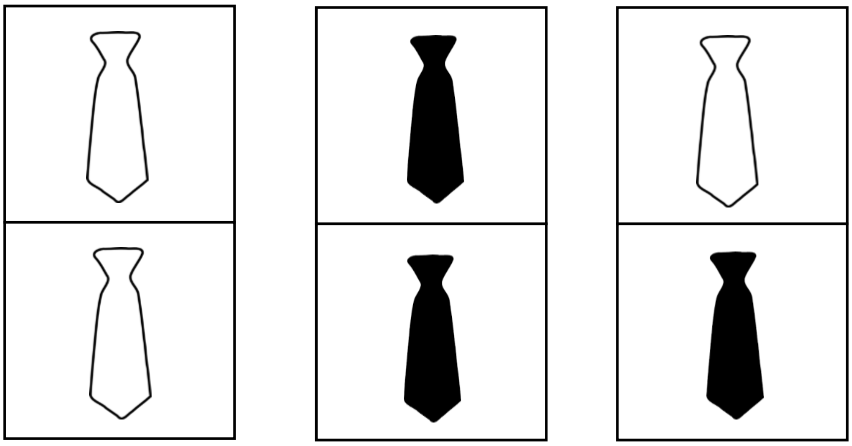
Question 19

Rearrange the digits in the rings so that the sum of digits inside each ring adds up to a total of 13.
What number will appear in the top centre ring?
Hint: Put the 7 in the top left circle, and put the 9 in the top right circle.



Question 20

There are three boxes, each with two compartments. Each compartment holds either a black tie or a white tie. The ties are placed in compartments and boxes exactly as depicted below.



You choose a box at random, then open a compartment at random. If the tie is black, what is the probability that the other tie in the box is also black?

Total: 200

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